Soil: account for the water

Teacher guide and worked answers

Use input minus collected water to estimate and compare water remaining after drainage.

40 minutes. Use Lesson.html for interactive teaching or Lesson.pptx for editable slides. Slides.pdf is the static alternative. Select one pupil response route and print the shared evidence it requires. Detailed answers below are for staff.

Scheme of work

_passsb/inputs/Build SOW 2026-2027.xlsx | BUILD Weekly - Spring | C32

Investigate which soil holds the most water; record results (ICT).

Direct match to C 32 using a carefully qualified classroom estimate: equal water input minus volume collected after five minutes. Results are recorded in tables; optional calculator/spreadsheet entry supports the ICT link. Limitations prevent claiming an exact soil-only water capacity.

Prior learning

Water can move through connected soil pores.

A fair comparison keeps input, collection time and sample preparation matched.

Subtracting an amount that leaves helps find an amount that is left.

Success evidence

Record input and collected volumes with mL units.

Subtract collected volume from input to find the amount not collected.

Compare samples and explain one reason the result is only an estimate of soil water held.

Equipment

Printed evidence page and selected response route; pencil and optional calculator.

Optional practical: identical draining plastic pots and filters, stable supports, spill tray, prepared soils A/B/C, 100 cm³ scoop, 60 mL water measure, graduated receivers and timer.

Optional ICT: a spreadsheet on a school device; enter the provided column headings and subtraction formula. No online account is needed.

Preparation

Pilot the method. Use 100 cm³ matched air-dry portions, loosely filled without tamping, identical pots/filters and 60 mL input poured over 10 seconds.

Use dry matching filters and initially dry pots/receivers. The prepared soil is approximately air-dry, not guaranteed water-free. Residual starting moisture makes this an estimate; the constructed model assumes negligible initial water.

Have the five-minute test ready to start early in the independent stage. One group can test one sample and share data; use supplied data for any missing samples.

Print one response route per pupil. Prepare a calculator if arithmetic might obscure the science.

If using ICT, create columns Soil, Input_mL, Collected_mL, Not_collected_mL. Pupils type =B2-C2 in D2 and copy down; paper is an equivalent route.

Practical care

Use teacher-screened soils from a known teaching source; avoid contaminated soil, visible mould, sharp items and dust clouds.

Keep soil in trays, use spoons and wash hands afterwards. Do not taste or smell closely.

Use stable plastic pots and containers over a spill tray; wipe spilled water promptly.

Use room-temperature water. An adult prepares drainage holes and checks supports; pupils do not cut containers.

Ready to teach alternative

Use the complete constructed dataset and water-accounting diagrams. All core questions can be answered on paper or in a locally created spreadsheet. The dataset assumes no spills, negligible evaporation and negligible initial water; practical results must be labelled separately.

Teaching sequence

Slide	Stage	Minutes	Focus
1	Opening	0-2	Where did the water go?
2	Retrieval	2-5	Three links from last time
3	I do	5-8	Account for the added water
4	I do	8-10	Be precise about “held”
5	I do	10-12	Compare equal water budgets
6	We do	12-16	The water-budget team
7	We do	16-20	Find the overclaim
8	Hinge	20-23	Which statement is sound?
9	You do	23-25	Investigate; record; explain
10	You do	Within stage	Keep the method matched
11	You do	25-35	Make a water account
12	You do	Within stage	Check the claim before sharing
13	Review	35-38	Lundy loop: improve the claim
14	Exit	38-40	Close the water account

Times are a suggested 40-minute route. Slides marked Within stage share the preceding stage allocation. The optional HTML timer starts only when selected.

1 Opening Where did the water go?

Recall the previous drainage comparison. Now the question is how much water has not left by a later, fixed time. We will estimate, then check what that number can and cannot mean.

2 Retrieval Three links from last time

Accept a diagram or pointed pore, then a control. Separate arithmetic support from conceptual understanding: a calculator is allowed. Briefly model 60−20 if needed.

Reveal: Connected pores. For example, equal input or time. 40.

3 I do Account for the added water

Think aloud:60 mL went in, 34 mL was collected, so 60−34=26 mL was not collected at five minutes. We assume no spills and negligible evaporation and negligible initial water. I am not yet saying all 26 mL is held inside the soil. The bar lengths accurately represent the constructed numbers.

4 I do Be precise about “held”

Trace each location. Say: input minus collected gives water not collected. With no losses this estimates water remaining in the whole setup. It is not an exact soil-only retention measurement or plant-available water. If standing water is visible, record it rather than calling it absorbed by soil.

5 I do Compare equal water budgets

Work one row together visually: A 46 out means 14 notcollected. B 34 out means 26. C 22 out means 38. Point to the same total bar lengths. The word left on the diagram is shorthand for not collected under the no-loss, negligible-initial-water assumption, not a promise that every drop is in soil pores.

Reveal: C:60−22=38 mL at five minutes, under the stated assumptions.

6 We do The water-budget team

Ask each pupil to choose an operation before the class explanation. Addition gives 94, which double-counts water rather than finding a remainder; 34−60 gives a negative amount and reverses input/output. Do not let arithmetic speed determine who contributes.

Reveal: 60−34=26 mL not collected. Input minus collected is the needed direction.

7 We do Find the overclaim

Choose the more careful claim first. P supports an approximate setup balance, still not soil-only. Q means some uncollected water is above the soil. R means some water was lost from the apparatus and not collected; repeat the trial, do not label it soil retention. Ask pupils to improve a claim in their own words.

Reveal: The second claim is measured. Q and R make a soil-only claim especially unreliable.

8 Hinge Which statement is sound?

If pupils choose A, return to the equal-length 60 mL bars and count what is left. If C, name one unmeasured factor such as root air or plant type. Then select an independent route.

A. A has the greater amount not collected. - A has 14 mL not collected; C has 38 mL. The large collected value means a smaller remainder.

B. C has the greater amount not collected at this time. - 60−22=38 mL, greater than A’s14 mL, under the stated conditions.

C. C will therefore grow every plant best. - The balance does not measure every plant need or how much water roots can use.

9 You do Investigate; record; explain

Distribute one route. Start an optional practical immediately so the five-minute collection fits inside the 12-minute independent stage. Use the same 60 mL input and 100 cm³ soil preparation as the model, if approved and piloted. Actual and constructed data must be clearly distinguished.

10 You do Keep the method matched

Use identical draining pots and matching filters. Loosely fill each with 100 cm³ of prepared soil, without tamping. Add 60 mL over 10 seconds. Start timing when pouring starts. Collect and read at the stated time. Use fresh equally prepared portions for repeats. Keep soil out of the collection scale. For this lesson read at 5 minutes. Check for water above soil, continued dripping and spills. Pupils record these observations; a spilled or overflowing trial cannot support the uncorrected remaining-water estimate.

11 You do Make a water account

For ICT, pupils enter headers Soil/Input_mL/Collected_mL/Not_collected_mL. Demonstrate =B2-C2 in D2, then fill down. Check against 60−46=14 so pupils understand the formula. The completed paper table is equally valid.

12 You do Check the claim before sharing

Shares independent time. Read the conclusion aloud if writing obscures meaning. Encourage “water not collected” or “estimated water remaining in the setup”, with no claim of exact plant-available water.

13 Review Lundy loop: improve the claim

Use one pupil’s caution - standing water, filter, spills or ongoing drainage - to improve the shared sentence. Name how their evidence changed the conclusion. Do not imply that no observation means no understanding.

Reveal: “C had the most water not collected at five minutes in this model test.”

14 Exit Close the water account

Core evidence is a correct supported subtraction and a sensible comparison. Recognising a limitation adds precision; if missing, use the apparatus diagram for a two-minute follow-up next lesson.

Reveal: 20 mL. Above the soil or in the filter. More water may drain later.

Answers and assessment

Retrieval 1-3

1 Connected spaces/pores between particles.2 Any relevant matched condition, input, time, soil volume, starting moisture, packing action, pot/filter or pour duration.3 60−20=40.

Worked water balance

B:60−34=26 mL not collected. Under no-spill, negligible-evaporation and negligible-initial-water assumptions this estimates water remaining in the whole setup, including any water in soil, on top or in the filter.

W1 / operation options

Choose 60−34=26 mL. 60+34=94 adds input and part of that same input, so does not find the remainder.34−60=−26 reverses the direction and is not a valid remainder here.

W2 / cards P, Q, R

The exact-soil claim is too strong because the filter may retain water and water may remain above soil. P supports a setup estimate only. Q shows one non-soil store. R means uncollected water includes a spill, so repeat or account for the measured loss; do not label it all as retained in soil.

Hinge

C has the greater amount not collected at this time:38 mL versus A 14 mL. A’s larger output means its remainder is smaller. A plant-growth conclusion goes beyond the measured balance.

S1-S5

S1 60−46. S2 14 mL. S3 C, 38 mL, greater than 26 and 14. S4 above the soil, in the filter or stuck to apparatus; a spill is a loss outside setup, not a retained store. S5 does not: some water is not inside soil, some soil water is held too tightly, and plant uptake was not measured. Accept one clear reason.

T1-T2

T1 identify constructed data or actual measurement; time 5 min from pour start; control 60 mL input, 100 cm^3 soil, 10-secondpour, matched moisture, loose packing action and identical pot/filter. T2 constructed:A 60/46/14, B 60/34/26, C 60/22/38 mL. Actual data may differ and must be used honestly; if output exceeds input, check starting wet apparatus or measurement before claiming negative retention.

T3-T5

T3 C 38>B 26>A 14 mL for sample data; actual data require their own comparisons. T4 "At 5 minutes, C had 38 mL not collected, the largest amount in this test; this estimates water remaining in the setup if no water was lost." T5 check for spills/leaks, standing water, filter storage, continued drips or substantial evaporation. Any one accurate limitation suffices.

X1-X4

X1 A 14, B 26, C 38 mL; C>B>A. X2 60−30=30 mL not collected; subtract 10 mL spill to estimate 20 mL remaining in the setup. X3 at 5 min 38 mL not collected; at 10 min 30 mL. An additional 8 mL drained between readings. X4 setup includes non-soil water; five/tenminutes is not field capacity; water held can be too tightly bound for roots; plant physiology and soil-air conditions are unmeasured. Accept a suitable concise explanation.

E1-E3

E1 20 mL. E2 filter, above-soil puddle, container surface. E3 more water can drain later, so the measured remainder depends on time; all comparisons need the same time.

R1-R2

Accept careful phrases such as "not collected", "atfive minutes", "in this setup" or "assuming no spills". R2 should be a relevant investigable question, e.g.what happens if drainage time increases; no fixed preferred question.

ICT

D2 formula=B2-C2. With the constructed data rows, A 14, B 26, C 38. Copying the formula down should adjust row references. A learner should check one result by hand or calculator rather than trust formatting alone.

Optional practical

Report actual input/output/time and visible standing water or losses. Input−collected is uncollected water; only under the no-loss approximation does it estimate what remains in the setup. For soil-only storage a different measurement method and correction for apparatus/water outside soil would be needed.

Teaching and assessment notes

40 minutes:opening 2, retrieval 3, I do 7, We do 8, hinge 3, You do 12, review 3, exit 2. The practical runs inside the 12-minute independent stage; zero-minute slides share that time.

Use the accessible phrase "water not collected" consistently. Input minus collected is a water balance, not automatically an exact measurement of water held by soil. With no spill/leak and negligible evaporation and negligible initial water, it approximates water remaining in the whole setup.

The SoW asks which soil holds most water. This lesson develops a useful classroom estimate while explicitly showing its limits. Match apparatus and preparation, record standing water and qualify the ranking. Do not award extra certainty for saying “C holds exactly 38 mL”.

The Week 4 and Week 5 constructed results are consistent:the same 60 mL input gives larger total collections by 5 minutes than by 2 minutes. They are authored datasets, not observed measurements or guaranteed values for actual soil types.

Teacher script: “I know what went in and what I caught. I subtract to find what I did not catch. Now I ask where it could be. Some may be in the soil, some elsewhere. That is why I call it an estimate.”

Organic matter and texture can affect water storage, but stored water is not identical to plant-available water. Avoid “clay holds most so clay is best” or “most water always means best growth”.

ICT is optional and offline: a spreadsheet is used to record measured/constructed data and calculate one transparent subtraction. The full table and method exist on paper, so no digital account or unavailable file is required.

If arithmetic is the barrier, use a calculator or completed remainder choices and assess the comparison and limitation. Pathway labels guide access, not fixed intelligence or diagnoses.

Initial water matters. If the setup begins wet, input minus collected estimates the increase in water in the setup, assuming negligible losses; it is not the total water present. Our constructed model assumes negligible starting water. Real air-dry soil can retain moisture, so qualify practical totals and record starting preparation.

Access and responsive teaching

Supported

Use a calculator, a60-unit strip or adult-scribed number sentence. Pupil selects the correct subtraction and compares completed values; the science remains the same.

Standard

Complete all three water-balance rows and an evidence claim, with one stated limitation. Optional spreadsheet has a taught formula.

Stretch

Evaluate a spill/error case and a longer-drainage result; distinguish the setup estimate from soil-only and plant-available water.

Regulation

Offer a dry recording role or paper task. A five-minute observation can include a quiet seated check-in. No compulsory contact with soil or timed arithmetic contest.

Misconceptions to check

The collection container with most water shows the soil that held most.

With equal input, more collected means less not collected at that time.

Check: Model 60 in and 46 out, then compare 60 in and 22 out.

Input minus collected is always exactly the water held in the soil.

Some may be above the soil or in the filter, and spills can escape collection. It is a setup estimate.

Check: Ask pupils to point to two possible locations outside the soil.

All retained water is available to plant roots.

Some water is held too tightly; roots also need air. This test does not measure plant-available water.

Check: Ask whether a waterlogged pot must grow every plant best.

The remainder is permanent.

Water can keep draining or evaporate, so the stated time matters.

Check: Ask what could happen if collection continues for another five minutes.

Word	Meaning
retain	Keep or hold.
input	The water we add.
collected	The water caught after it drains out.
estimate	An approximate value based on measurements and assumptions.
water balance	Accounting for where the added water goes.

Scientific references

USDA NRCS: Soil Bulk Density, Moisture and Aeration — Guides for Educators

Checked 8 September 2026. Supports pore space, stored water, organic matter and the distinction between water held and water available to plants. Short-time classroom water balance is an estimate and not a field-capacity measurement.

USDA NRCS: Soil Infiltration — Guides for Educators

Checked 8 September 2026. Supports effects of pores, texture, initial moisture and compaction on water movement. Our pot drainage test is a comparative classroom model, not a measurement of field infiltration or intrinsic permeability.